

# MACHINE LEARNING

## HOMEWORK – 5

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### Problem – 1

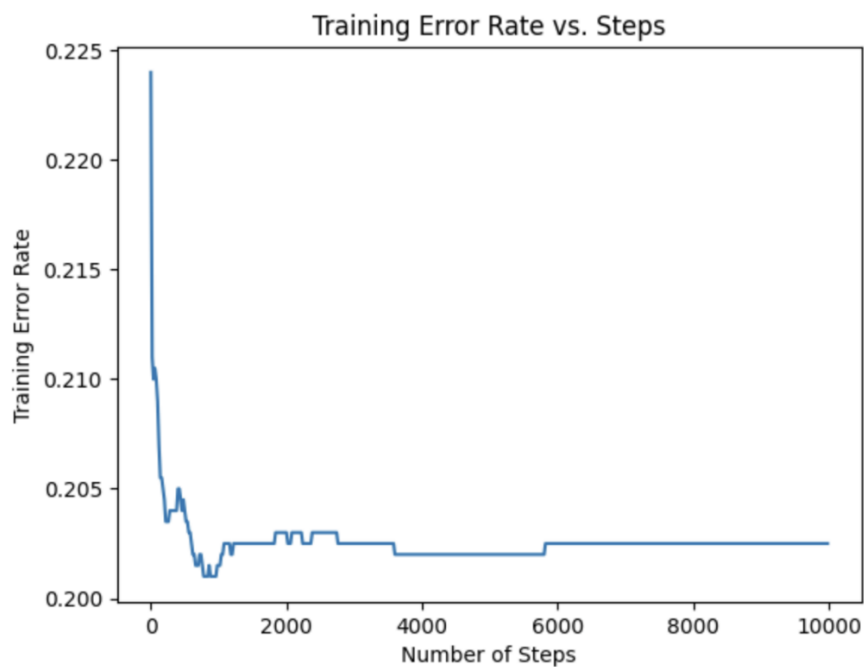
A.

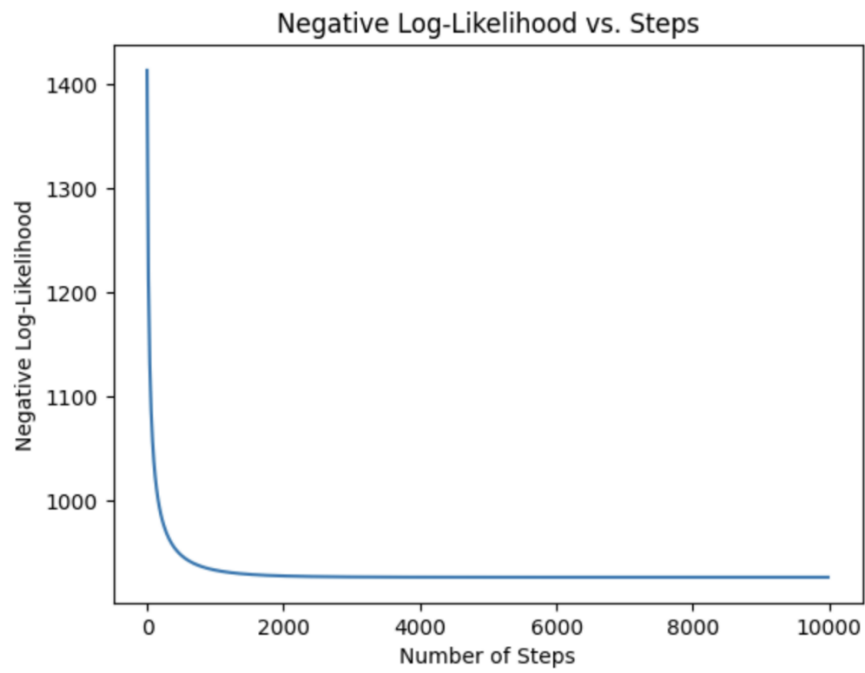
**Training Error Rate:** 0.2025

**Test Error Rate:** 0.2092

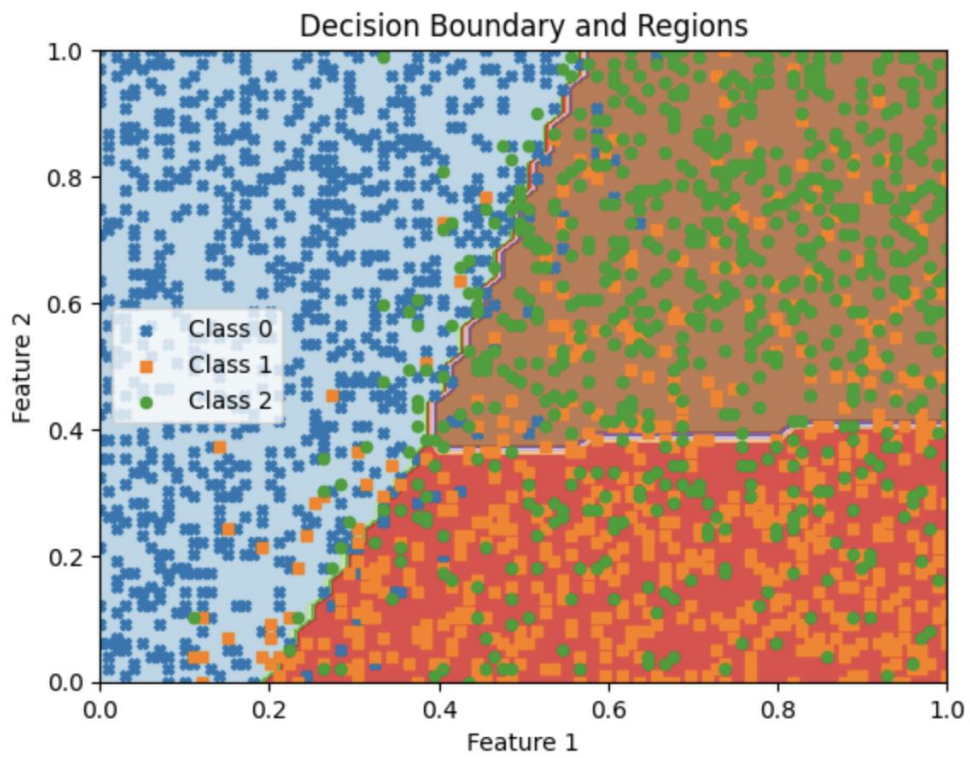
B.

We noticed that they both decay at something that looks like a polynomial rate but it's interesting because while the Negative Log Likelihood decays smoothly the training error rate jumps around a little bit even going through a minimum around 800 steps.





Just for fun here is the decision boundary produced by the gradient descent method:



## Problem – 2

A.

We know,

$$J: R^2 \rightarrow R$$

$$J(w) = \frac{1}{2}w_1^2 + \ln(1 + \exp(2w_1 + w_2)) - (2w_1 + w_2) \\ + \ln(1 + \exp(-2w_1 + w_2))$$

Let

$$J_1(w) = \frac{1}{2}w_1^2$$

$$J_2(w) = \ln(1 + \exp(2w_1 + w_2))$$

$$J_3(w) = -(2w_1 + w_2)$$

$$J_4(w) = \ln(1 + \exp(-2w_1 + w_2))$$

Therefore,

$$J(w) = J_1(w) + J_2(w) + J_3(w) + J_4(w)$$

→ Consider  $J_1(w)$

$$J_1(w) = \frac{1}{2}w_1^2$$

Computing double derivative for  $J_1(w)$ ,

$$\frac{\partial J_1}{\partial w_1} = w_1$$

$$\frac{\partial^2 J_1}{\partial w_1^2} = 1$$

$$\frac{\partial J_1}{\partial w_2} = 0$$

$$\frac{\partial^2 J_1}{\partial w_2^2} = 0$$

$$\frac{\partial^2 J_1}{\partial w_1 \partial w_2} = 0$$

$$\frac{\partial^2 J_1}{\partial w_2 \partial w_1} = 0$$

Constructing the Hessian matrix:

$$H = \begin{pmatrix} \frac{\partial^2 J_1}{\partial w_1^2} & \frac{\partial^2 J_1}{\partial w_1 \partial w_2} \\ \frac{\partial^2 J_1}{\partial w_2 \partial w_1} & \frac{\partial^2 J_1}{\partial w_2^2} \end{pmatrix}$$

$$H = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

The dimensions of Hessian matrix is 2 X 2

Analysing the positive semidefinite-ness of H:

For any vector  $v$ , such that  $v \in R$ , let

$$v = (v_1 \quad v_2)$$

Thus,

$$v^T H v = v_1^2$$

$$\Rightarrow v^T H v \geq 0$$

Therefore,  $v^T H v$  is always non-negative.

$\Rightarrow$  The Hessian matrix H is positive semi-definite.

**This implies that  $J_1(w)$  is a convex function. (1)**

➔ Consider  $J_2(w)$

$$J_2(w) = \ln(1 + \exp(2w_1 + w_2))$$

Computing derivatives,

$$\frac{\partial J_2}{\partial w_1} = \frac{2 \exp(2w_1 + w_2)}{1 + \exp(2w_1 + w_2)}$$

$$\frac{\partial J_2}{\partial w_2} = \frac{\exp(2w_1 + w_2)}{1 + \exp(2w_1 + w_2)}$$

Computing the double derivatives for  $J_2(w)$ ,

$$\frac{\partial^2 J_2}{\partial w_1^2} = \frac{4 \exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2}$$

$$\frac{\partial^2 J_2}{\partial w_2^2} = \frac{\exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2}$$

$$\frac{\partial^2 J_2}{\partial w_1 \partial w_2} = \frac{2 \exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2}$$

$$\frac{\partial^2 J_2}{\partial w_2 \partial w_1} = \frac{2 \exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2}$$

Constructing the Hessian matrix,

$$H = \begin{pmatrix} \frac{\partial^2 J_2}{\partial w_1^2} & \frac{\partial^2 J_2}{\partial w_1 \partial w_2} \\ \frac{\partial^2 J_2}{\partial w_2 \partial w_1} & \frac{\partial^2 J_2}{\partial w_2^2} \end{pmatrix}$$

$$H = \begin{pmatrix} \frac{4 \exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2} & \frac{2 \exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2} \\ \frac{2 \exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2} & \frac{\exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2} \end{pmatrix}$$

The Hessian matrix here is of dimensions 2 X 2

Analysing the positive semi-definiteness of H,

For any vector  $v$ , such that  $v \in R^2$

Let  $v = (v_1 \quad v_2)$

Thus,

$$v^T H v = \frac{4v_1^2 \exp(2w_1 + w_2) + 4v_1 v_2 \exp(2w_1 + w_2) + v_2^2 \exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2}$$

$$\Rightarrow v^T H v = \frac{\exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2} (4v_1^2 + 4v_1 v_2 + v_2^2)$$

$$\Rightarrow v^T H v = \frac{\exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2} (2v_1 + v_2)^2$$

Thus,  $(2v_1 + v_2)^2$  is always non-negative, and  $\frac{\exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2}$  is also always non-negative.

Therefore,

$$(2v_1 + v_2)^2 \geq 0$$

$$\frac{\exp(2w_1 + w_2)}{(1 + \exp(2w_1 + w_2))^2} \geq 0$$

$$\Rightarrow v^T H v \geq 0$$

Therefore,  $v^T H v$  is always non-negative.

$\Rightarrow$  The Hessian matrix  $H$  is positive semi-definite.

**This implies that  $J_2(w)$  is a convex function (2)**

$\rightarrow$  Consider  $J_3(w)$

$$J_3(w) = -(2w_1 + w_2)$$

Computing derivatives,

$$\frac{\partial J_3}{\partial w_1} = -2$$

$$\frac{\partial J_3}{\partial w_2} = -1$$

Computing the double derivatives for  $J_3(w)$ ,

$$\frac{\partial^2 J_3}{\partial w_1^2} = 0$$

$$\frac{\partial^2 J_3}{\partial w_2^2} = 0$$

$$\frac{\partial^2 J_3}{\partial w_1 \partial w_2} = 0$$

$$\frac{\partial^2 J_3}{\partial w_2 \partial w_1} = 0$$

Constructing the Hessian matrix,

$$H = \begin{pmatrix} \frac{\partial^2 J_3}{\partial w_1^2} & \frac{\partial^2 J_3}{\partial w_1 \partial w_2} \\ \frac{\partial^2 J_3}{\partial w_2 \partial w_1} & \frac{\partial^2 J_3}{\partial w_2^2} \end{pmatrix}$$

$$H = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

The Hessian matrix here is of dimensions 2 X 2

Analysing the positive semi-definiteness of H,

For any vector  $v$ , such that  $v \in R^2$

Let  $v = (v_1 \quad v_2)$

Thus,

$$v^T H v = 0$$

$$\Rightarrow v^T H v \geq 0$$

Therefore,  $v^T H v$  is always non-negative.

$\Rightarrow$  The Hessian matrix H is positive semi-definite.

**This implies that  $J_3(w)$  is a convex function. (3)**

**$\rightarrow$  Consider  $J_4(w)$**

$$J_4(w) = \ln(1 + \exp(-2w_1 + w_2))$$

Computing derivatives,

$$\frac{\partial J_4}{\partial w_1} = \frac{-2 \exp(-2w_1 + w_2)}{1 + \exp(-2w_1 + w_2)}$$

$$\frac{\partial J_4}{\partial w_2} = \frac{\exp(-2w_1 + w_2)}{1 + \exp(-2w_1 + w_2)}$$

Computing the double derivatives for  $J_4(w)$ ,

$$\frac{\partial^2 J_4}{\partial w_1^2} = \frac{4 \exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2}$$



$$\frac{\partial^2 J_4}{\partial w_2^2} = \frac{\exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2}$$

$$\frac{\partial^2 J_4}{\partial w_1 \partial w_2} = \frac{-2 \exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2}$$

$$\frac{\partial^2 J_4}{\partial w_2 \partial w_1} = \frac{-2 \exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2}$$

Constructing the Hessian matrix,

$$H = \begin{pmatrix} \frac{\partial^2 J_4}{\partial w_1^2} & \frac{\partial^2 J_4}{\partial w_1 \partial w_2} \\ \frac{\partial^2 J_4}{\partial w_2 \partial w_1} & \frac{\partial^2 J_4}{\partial w_2^2} \end{pmatrix}$$

$$H = \begin{pmatrix} \frac{4 \exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2} & \frac{-2 \exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2} \\ \frac{-2 \exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2} & \frac{\exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2} \end{pmatrix}$$

The Hessian matrix here is of dimensions 2 X 2

Analysing the positive semi-definiteness of H,

For any vector  $v$ , such that  $v \in R^2$

Let  $v = (v_1 \quad v_2)$

Thus,

$$v^T H v$$

$$= \frac{4v_1^2 \exp(-2w_1 + w_2) - 4v_1 v_2 \exp(-2w_1 + w_2) + v_2^2 \exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2}$$

$$\Rightarrow v^T H v = \frac{\exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2} (4v_1^2 - 4v_1 v_2 + v_2^2)$$

$$\Rightarrow v^T H v = \frac{\exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2} (2v_1 - v_2)^2$$

Thus,  $(2v_1 - v_2)^2$  is always non-negative, and  $\frac{\exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2}$  is also always non-negative.

Therefore,

$$(2v_1 - v_2)^2 \geq 0$$

$$\frac{\exp(-2w_1 + w_2)}{(1 + \exp(-2w_1 + w_2))^2} \geq 0$$

$$\Rightarrow v^T H v \geq 0$$

Therefore,  $v^T H v$  is always non-negative.

$\Rightarrow$  The Hessian matrix  $H$  is positive semi-definite.

**This implies that  $J_4(w)$  is a convex function** (4)

Using (1), (2), (3) and (4),

**$J_1(w), J_2(w), J_3(w),$  and  $J_4(w)$  are convex functions.**

Therefore,

$$J(w) = J_1(w) + J_2(w) + J_3(w) + J_4(w)$$

Hence, we know that the sum operation preserves convexity.

Therefore, as  $J_1(w), J_2(w), J_3(w),$  and  $J_4(w)$  are convex functions,  $J(w)$  is also a convex function.

**$\Rightarrow J(w)$  is a convex function.**

## B.

We know,

$$J: R^2 \rightarrow R$$

$$J(w) = \frac{1}{2}w_1^2 + \ln(1 + \exp(2w_1 + w_2)) - (2w_1 + w_2) \\ + \ln(1 + \exp(-2w_1 + w_2))$$

Gradient of  $J$  is given by:

$$\nabla J = \begin{pmatrix} \frac{\partial J}{\partial w_1} \\ \frac{\partial J}{\partial w_2} \end{pmatrix}$$

Computing the partial derivatives:

1.  $\frac{\partial J}{\partial w_1}$

$$\frac{\partial J}{\partial w_1} =$$

$$w_1 + \frac{1}{1+\exp(2w_1+w_2)} \left( \frac{\partial(\exp(2w_1+w_2))}{\partial w_1} \right) - 2 + \frac{1}{1+\exp(-2w_1+w_2)} \left( \frac{\partial(\exp(-2w_1+w_2))}{\partial w_1} \right)$$

$$\Rightarrow \frac{\partial J}{\partial w_1} = w_1 + \frac{\exp(2w_1+w_2)}{1+\exp(2w_1+w_2)} \left( \frac{\partial(2w_1+w_2)}{\partial w_1} \right) - 2 + \\ \frac{\exp(-2w_1+w_2)}{1+\exp(-2w_1+w_2)} \left( \frac{\partial(-2w_1+w_2)}{\partial w_1} \right)$$

$$\Rightarrow \frac{\partial J}{\partial w_1} = w_1 + \frac{2 \exp(2w_1+w_2)}{1+\exp(2w_1+w_2)} - 2 - \frac{2 \exp(-2w_1+w_2)}{1+\exp(-2w_1+w_2)} \quad (1)$$

2.  $\frac{\partial J}{\partial w_2}$

$$\frac{\partial J}{\partial w_2} = \frac{1}{1+\exp(2w_1+w_2)} \left( \frac{\partial(\exp(2w_1+w_2))}{\partial w_2} \right) - 1 + \frac{1}{1+\exp(-2w_1+w_2)} \left( \frac{\partial(\exp(-2w_1+w_2))}{\partial w_2} \right)$$

$$\Rightarrow \frac{\partial J}{\partial w_2} = \frac{\exp(2w_1 + w_2)}{1 + \exp(2w_1 + w_2)} \left( \frac{\partial(2w_1 + w_2)}{\partial w_2} \right) - 1 + \frac{\exp(-2w_1 + w_2)}{1 + \exp(-2w_1 + w_2)} \left( \frac{\partial(-2w_1 + w_2)}{\partial w_2} \right)$$

$$\Rightarrow \frac{\partial J}{\partial w_2} = \frac{\exp(2w_1 + w_2)}{1 + \exp(2w_1 + w_2)} - 1 + \frac{\exp(-2w_1 + w_2)}{1 + \exp(-2w_1 + w_2)} \quad (2)$$

Now,

$$\nabla J = \begin{pmatrix} \frac{\partial J}{\partial w_1} \\ \frac{\partial J}{\partial w_2} \end{pmatrix}$$

Using (1) and (2),

$$\Rightarrow \nabla J = \begin{pmatrix} w_1 + \frac{2 \exp(2w_1 + w_2)}{1 + \exp(2w_1 + w_2)} - 2 - \frac{2 \exp(-2w_1 + w_2)}{1 + \exp(-2w_1 + w_2)} \\ \frac{\exp(2w_1 + w_2)}{1 + \exp(2w_1 + w_2)} - 1 + \frac{\exp(-2w_1 + w_2)}{1 + \exp(-2w_1 + w_2)} \end{pmatrix}$$

### Problem – 3

A.

We know,

$$J(w) = (\text{logistic}(2w_1 + w_2) - 1)^2 + (\text{logistic}(-2w_1 + w_2) - 0)^2$$

$$\Rightarrow J(w) = (\text{logistic}(2w_1 + w_2) - 1)^2 + (\text{logistic}(-2w_1 + w_2))^2$$

Let,

$$J_1(w) = (\text{logistic}(2w_1 + w_2) - 1)^2$$

$$J_2(w) = (\text{logistic}(-2w_1 + w_2))^2$$

$$\Rightarrow J(w) = J_1(w) + J_2(w)$$

→

Consider  $J_1(w)$ :

$$J_1(w) = (\text{logistic}(2w_1 + w_2) - 1)^2$$

For simplicity in computing derivatives, let:

$$z = 2w_1 + w_2$$

$$g(z) = \text{logistic}(z)$$

$$\Rightarrow J_1 = (g(z) - 1)^2$$

Also, we know:

$$g'(z) = \frac{\partial g(z)}{\partial z}$$

$$\Rightarrow g'(z) = \frac{\partial(\text{logistic}(z))}{\partial z}$$

$$\Rightarrow g'(z) = \frac{\partial}{\partial z} \left( \frac{1}{1 + e^{-z}} \right)$$

$$\Rightarrow g'(z) = \frac{e^{-z}}{(1 + e^{-z})^2}$$

$$\Rightarrow g'(z) = \text{logistic}(z)(1 - \text{logistic}(z))$$

$$\Rightarrow g'(z) = g(z)(1 - g(z))$$

Computing the derivatives for  $J_1$ :

$$\frac{\partial J_1}{\partial w_1} = 2(g(z) - 1) \frac{\partial z}{\partial w_1} g'(z) \frac{\partial z}{\partial w_1}$$

$$\Rightarrow \frac{\partial J_1}{\partial w_1} = 2(g(z) - 1)(2)g'(z)(2)$$

$$\Rightarrow \frac{\partial J_1}{\partial w_1} = 8(g(z) - 1)g'(z)$$

$$\Rightarrow \frac{\partial J_1}{\partial w_1} = -8g(z)(1 - g(z))^2$$

$$\frac{\partial J_1}{\partial w_2} = 2(g(z) - 1) \frac{\partial z}{\partial w_2} g'(z) \frac{\partial z}{\partial w_2}$$

$$\Rightarrow \frac{\partial J_1}{\partial w_2} = 2(g(z) - 1)(1)g'(z)(1)$$

$$\Rightarrow \frac{\partial J_1}{\partial w_2} = 2(g(z) - 1)g'(z)$$

$$\Rightarrow \frac{\partial J_1}{\partial w_2} = -2g(z)(1 - g(z))^2$$

Now,

Computing the second derivatives for  $J_1(w)$ ,

$$\frac{\partial^2 J_1}{\partial w_1^2} = \frac{\partial}{\partial w_1} \left( -8g(z)(1 - g(z))^2 \right)$$

$$\Rightarrow \frac{\partial^2 J_1}{\partial w_1^2} = -8 \left( (1 - g(z))^2 \frac{\partial}{\partial w_1} (g(z)) + \frac{\partial}{\partial w_1} ((1 - g(z))^2) \right)$$

$$\Rightarrow \frac{\partial^2 J_1}{\partial w_1^2} = -8 \left( (1 - g(z))^2 g'(z) \frac{\partial z}{\partial w_1} + g(z) \frac{\partial}{\partial w_1} ((1 - g(z))^2) \right)$$

$$\Rightarrow \frac{\partial^2 J_1}{\partial w_1^2} = -8 \left( 2g'(z)(1 - g(z))^2 + g(z) \frac{\partial}{\partial w_1} ((1 - g(z))^2) \right)$$

Now, we see that:

$$J_1 = (g(z) - 1)^2 = (1 - g(z))^2$$

So,

$$\Rightarrow \frac{\partial^2 J_1}{\partial w_1^2} = -8 \left( 2g'(z)(1 - g(z))^2 + g(z) \frac{\partial J_1}{\partial w_1} \right)$$

$$\Rightarrow \frac{\partial^2 J_1}{\partial w_1^2} = -8 \left( 2g(z)(1 - g(z))^3 - 8(g(z))^2(1 - g(z))^2 \right)$$

$$\Rightarrow \frac{\partial^2 J_1}{\partial w_1^2} = -16g(z)(1 - g(z))^2(1 - g(z) - 4g(z))$$

$$\Rightarrow \frac{\partial^2 J_1}{\partial w_1^2} = -16g(z)(1 - g(z))^2(1 - 5g(z))$$

$$\begin{aligned}
\frac{\partial^2 J_1}{\partial w_2^2} &= \frac{\partial}{\partial w_2} \left( -2g(z)(1-g(z))^2 \right) \\
&\Rightarrow \frac{\partial^2 J_1}{\partial w_2^2} = -2 \frac{\partial}{\partial w_2} \left( g(z)(1-g(z))^2 \right) \\
&\Rightarrow \frac{\partial^2 J_1}{\partial w_2^2} = -2 \left( (1-g(z))^2 \frac{\partial}{\partial w_2} (g(z)) + g(z) \frac{\partial}{\partial w_2} ((1-g(z))^2) \right)
\end{aligned}$$

Like how I computed before,

$$\begin{aligned}
&\Rightarrow \frac{\partial^2 J_1}{\partial w_2^2} = -2 \left( g'(z)(1-g(z))^2 + g(z) \frac{\partial J_1}{\partial w_2} \right) \\
&\Rightarrow \frac{\partial^2 J_1}{\partial w_2^2} = -2 \left( g'(z)(1-g(z))^2 - 2(g(z))^2(1-g(z))^2 \right) \\
&\Rightarrow \frac{\partial^2 J_1}{\partial w_2^2} = -2 \left( g(z)(1-g(z))(1-g(z))^2 - 2(g(z))^2(1-g(z))^2 \right) \\
&\Rightarrow \frac{\partial^2 J_1}{\partial w_2^2} = -2 \left( g(z)(1-g(z))^3 - 2(g(z))^2(1-g(z))^2 \right) \\
&\Rightarrow \frac{\partial^2 J_1}{\partial w_2^2} = -2g(z)(1-g(z))^2(1-g(z)-2g(z)) \\
&\Rightarrow \frac{\partial^2 J_1}{\partial w_2^2} = -2g(z)(1-g(z))^2(1-3g(z))
\end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 J_1}{\partial w_2 \partial w_1} &= \frac{\partial}{\partial w_2} \left( -8g(z)(1-g(z))^2 \right) \\
&\Rightarrow \frac{\partial^2 J_1}{\partial w_2 \partial w_1} = -8 \frac{\partial}{\partial w_2} \left( g(z)(1-g(z))^2 \right)
\end{aligned}$$

Already computed this partial derivative before,

$$\begin{aligned}
&\Rightarrow \frac{\partial^2 J_1}{\partial w_2 \partial w_1} = -8 \left( g'(z)(1-g(z))^2 - 2(g(z))^2(1-g(z))^2 \right) \\
&\Rightarrow \frac{\partial^2 J_1}{\partial w_2 \partial w_1} = -8 \left( g(z)(1-g(z))^3 - 2(g(z))^2(1-g(z))^2 \right) \\
&\Rightarrow \frac{\partial^2 J_1}{\partial w_2 \partial w_1} = -8g(z)(1-g(z))(1-3g(z))
\end{aligned}$$

$$\frac{\partial^2 J_1}{\partial w_1 \partial w_2} = \frac{\partial}{\partial w_1} \left( -2g(z)(1-g(z))^2 \right)$$

$$\Rightarrow \frac{\partial^2 J_1}{\partial w_1 \partial w_2} = -2 \frac{\partial}{\partial w_1} \left( g(z)(1-g(z))^2 \right)$$

Already computed this partial derivative before,

$$\Rightarrow \frac{\partial^2 J_1}{\partial w_1 \partial w_2} = -2 \left( 2g(z)(1-g(z))^3 - 8(g(z))^2(1-g(z))^2 \right)$$

$$\Rightarrow \frac{\partial^2 J_1}{\partial w_1 \partial w_2} = -4 \left( g(z)(1-g(z))^3 - 4(g(z))^2(1-g(z))^2 \right)$$

$$\Rightarrow \frac{\partial^2 J_1}{\partial w_1 \partial w_2} = -4g(z)(1-g(z))^2(1-g(z)-4g(z))$$

$$\Rightarrow \frac{\partial^2 J_1}{\partial w_1 \partial w_2} = -4g(z)(1-g(z))^2(1-5g(z))$$

Constructing the Hessian matrix for  $J_1$ :

$$H = \begin{pmatrix} \frac{\partial^2 J_1}{\partial w_1^2} & \frac{\partial^2 J_1}{\partial w_1 \partial w_2} \\ \frac{\partial^2 J_1}{\partial w_2 \partial w_1} & \frac{\partial^2 J_1}{\partial w_2^2} \end{pmatrix}$$

$$H = \begin{pmatrix} -16g(z)(1-g(z))^2(1-5g(z)) & -4g(z)(1-g(z))^2(1-5g(z)) \\ -8g(z)(1-g(z))(1-3g(z)) & -2g(z)(1-g(z))^2(1-3g(z)) \end{pmatrix}$$

The Hessian matrix here is of dimensions 2 X 2

Analysing the positive semi-definiteness of H,

For any vector  $v$ , such that  $v \in R^2$

Let  $v = (v_1 \quad v_2)$

Thus,

$$\begin{aligned} v^T H v &= -16v_1^2 g(z)(1-g(z))^2(1-5g(z)) - 8v_1 v_2 g(z)(1-g(z))(1-3g(z)) \\ &\quad - 4v_1 v_2 g(z)(1-g(z))^2(1-5g(z)) - 2v_2^2 g(z)(1-g(z))^2(1-3g(z)) \end{aligned}$$



$$v^T H v = g(z)(1 - g(z))^2 \left( -16v_1^2(1 - 5g(z)) - 8v_1v_2(1 - 3g(z)) - 4v_1v_2(1 - 5g(z)) - 2v_2^2(1 - 3g(z)) \right)$$

$$v^T H v = g(z)(1 - g(z))^2 \left( (-16v_1^2 - 12v_1v_2 + 2v_2^2) + g(z)(80v_1^2 + 44v_1v_2 + 6v_2^2) \right)$$

$$v^T H v = g(z)(1 - g(z))^2 \left( -(8v_1 + v_2)(2v_1 + v_2) + 16g(z)(v_1 + 10v_2)(5v_1 + 3v_2) \right)$$

Now,  $g(z)$  and  $(1 - g(z))^2$  are always non-negative

$$g(z) \geq 0$$

$$(1 - g(z))^2 \geq 0$$

However, the term  $-(8v_1 + v_2)(2v_1 + v_2) + 16g(z)(v_1 + 10v_2)(5v_1 + 3v_2)$  may be negative.

Let us consider the following set of values:

$$g(z) = 0.5, v_1 = -1, v_2 = 1,$$

For these set of values,

$$(1 - g(z))^2 = (1 - 0.5)^2 = 0.25$$

$$\begin{aligned} & (-(8v_1 + v_2)(2v_1 + v_2) + 16g(z)(v_1 + 10v_2)(5v_1 + 3v_2)) \\ &= \left( -(8(-1) + 1)(2(-1) + 1) + 16(0.5)(-1 + 10(1))(5(-1) + 3(1)) \right) \end{aligned}$$

$$\Rightarrow (-(8v_1 + v_2)(2v_1 + v_2) + 16g(z)(v_1 + 10v_2)(5v_1 + 3v_2)) = (-7 + 16(0.5)(-18))$$

$$\Rightarrow (-(8v_1 + v_2)(2v_1 + v_2) + 16g(z)(v_1 + 10v_2)(5v_1 + 3v_2)) = -151$$

$$\Rightarrow (-(8v_1 + v_2)(2v_1 + v_2) + 16g(z)(v_1 + 10v_2)(5v_1 + 3v_2)) \text{ is negative}$$

Hence,

$$v^T H v = 0.5(0.25)(-151)$$

$$\Rightarrow v^T H v = -18.875$$

$$\Rightarrow v^T H v < 0$$

Therefore,  $v^T H v$  is not always non-negative.

$\Rightarrow$  The Hessian matrix  $H$  is not positive semi-definite.

**This implies that  $J_1(w)$  is not a convex function.**

→

Consider  $J_2(w)$ :

$$J_2(w) = (\text{logistic}(-2w_1 + w_2))^2$$

For simplicity in computing derivatives, let:

$$z = -2w_1 + w_2$$

$$g(z) = \text{logistic}(z)$$

$$\Rightarrow J_2 = (g(z))^2$$

Also, we know:

$$g'(z) = \frac{\partial g(z)}{\partial z}$$

$$\Rightarrow g'(z) = \frac{\partial(\text{logistic}(z))}{\partial z}$$

$$\Rightarrow g'(z) = \frac{\partial}{\partial z} \left( \frac{1}{1 + e^{-z}} \right)$$

$$\Rightarrow g'(z) = \frac{e^{-z}}{(1 + e^{-z})^2}$$

$$\Rightarrow g'(z) = \text{logistic}(z)(1 - \text{logistic}(z))$$

$$\Rightarrow g'(z) = g(z)(1 - g(z))$$

Computing the derivatives for  $J_2$ :

$$\frac{\partial J_2}{\partial w_1} = 2g(z) \frac{\partial z}{\partial w_1} g'(z) \frac{\partial z}{\partial w_1}$$

$$\frac{\partial J_2}{\partial w_1} = 2g(z)(-2)g'(z)(-2)$$

$$\frac{\partial J_2}{\partial w_1} = 8g(z)g'(z)$$

$$\frac{\partial J_2}{\partial w_2} = 2g(z) \frac{\partial z}{\partial w_2} g'(z) \frac{\partial z}{\partial w_2}$$

$$\frac{\partial J_2}{\partial w_2} = 2g(z)(1)g'(z)(1)$$

$$\frac{\partial J_2}{\partial w_2} = 2g(z)g'(z)$$

Now,

$$g''(z) = \frac{\partial^2 g(z)}{\partial z^2}$$

$$\Rightarrow g''(z) = \frac{\partial g'(z)}{\partial z}$$

$$\Rightarrow g''(z) = \frac{\partial}{\partial z} (g(z)(1 - g(z)))$$

$$\Rightarrow g''(z) = (1 - g(z)) \frac{\partial}{\partial z} (g(z)) + g(z) \frac{\partial}{\partial z} (1 - g(z))$$

$$\Rightarrow g''(z) = (1 - g(z))g'(z) - g(z)g'(z)$$

$$\Rightarrow g''(z) = g'(z)(1 - 2g(z))$$

Computing second derivatives for  $J_2(w)$ :

$$\frac{\partial^2 J_2}{\partial w_1^2} = \frac{\partial}{\partial w_1} (8g(z)g'(z))$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_1^2} = 8g'(z) \frac{\partial}{\partial w_1} (g(z)) + 8g(z) \frac{\partial}{\partial w_1} (g'(z))$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_1^2} = 8(g'(z))^2 \frac{\partial z}{\partial w_1} + 8g(z)g''(z) \frac{\partial z}{\partial w_1}$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_1^2} = -16 \left( (g'(z))^2 + g(z)g'(z)(1 - 2g(z)) \right)$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_1^2} = -16g'(z) (g'(z) + g(z)(1 - 2g(z)))$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_1^2} = -16g'(z) (g'(z) + g(z)(1 - 2g(z)))$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_1^2} = -16g'(z) (g(z)(1 - g(z)) + g(z)(1 - 2g(z)))$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_1^2} = -16g'(z)g(z) ((1 - g(z)) + (1 - 2g(z)))$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_1^2} = -16g'(z)g(z)(2 - 3g(z))$$

$$\frac{\partial^2 J_2}{\partial w_2^2} = \frac{\partial}{\partial w_2} (2g(z)g'(z))$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_2^2} = 2g'(z) \frac{\partial}{\partial w_2} (g(z)) + 2g(z) \frac{\partial}{\partial w_2} (g'(z))$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_2^2} = 2(g'(z))^2 \frac{\partial z}{\partial w_2} + 2g(z)g''(z) \frac{\partial z}{\partial w_2}$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_2^2} = 2 \left( (g'(z))^2 + g(z)g'(z)(1 - 2g(z)) \right)$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_2^2} = 2g'(z) \left( g(z)(1 - g(z)) + g(z)(1 - 2g(z)) \right)$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_2^2} = 2g'(z)g(z) \left( (1 - g(z)) + (1 - 2g(z)) \right)$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_2^2} = 2g'(z)g(z)(2 - 3g(z))$$

$$\frac{\partial^2 J_2}{\partial w_1 \partial w_2} = \frac{\partial}{\partial w_1} (2g(z)g'(z))$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_1 \partial w_2} = 2(g'(z))^2 \frac{\partial z}{\partial w_1} + 2g(z)g''(z) \frac{\partial z}{\partial w_1}$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_1 \partial w_2} = -4 \left( (g'(z))^2 + g(z)g'(z)(1 - 2g(z)) \right)$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_1 \partial w_2} = -4g(z)g'(z) \left( (1 - g(z)) + (1 - 2g(z)) \right)$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_1 \partial w_2} = -4g'(z)g(z)(2 - 3g(z))$$

$$\frac{\partial^2 J_2}{\partial w_2 \partial w_1} = \frac{\partial}{\partial w_1} (8g(z)g'(z))$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_2 \partial w_1} = 8(g'(z))^2 \frac{\partial z}{\partial w_1} + 8g(z)g''(z) \frac{\partial z}{\partial w_1}$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_2 \partial w_1} = 8 \left( (g'(z))^2 + g(z)g'(z)(1 - 2g(z)) \right)$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_2 \partial w_1} = 8g(z)g'(z) \left( (1 - g(z)) + (1 - 2g(z)) \right)$$

$$\Rightarrow \frac{\partial^2 J_2}{\partial w_2 \partial w_1} = 8g'(z)g(z)(2 - 3g(z))$$

Constructing the Hessian matrix:

$$H = \begin{pmatrix} \frac{\partial^2 J_2}{\partial w_1^2} & \frac{\partial^2 J_2}{\partial w_1 \partial w_2} \\ \frac{\partial^2 J_2}{\partial w_2 \partial w_1} & \frac{\partial^2 J_2}{\partial w_2^2} \end{pmatrix}$$

$$H = \begin{pmatrix} -16g'(z)g(z)(2 - 3g(z)) & -4g'(z)g(z)(2 - 3g(z)) \\ 8g'(z)g(z)(2 - 3g(z)) & 2g'(z)g(z)(2 - 3g(z)) \end{pmatrix}$$

The Hessian matrix here is of dimensions 2 X 2

Analysing the positive semi-definiteness of H,

For any vector  $v$ , such that  $v \in R^2$

Let  $v = (v_1 \quad v_2)$

Thus,

$$v^T H v = -16v_1^2 g'(z)g(z)(2 - 3g(z)) + 4v_1v_2 g'(z)g(z)(2 - 3g(z)) + 2v_2^2 g'(z)g(z)(2 - 3g(z))$$

$$v^T H v = (-16v_1^2 + 4v_1v_2 + 2v_2^2) g'(z)g(z)(2 - 3g(z))$$

$$v^T H v = (-16v_1^2 + 4v_1v_2 + 2v_2^2) g'(z)g(z)(2 - 3g(z))$$

$$v^T H v = (-8v_1 - 2v_2)(2v_1 - v_2) g'(z)g(z)(2 - 3g(z))$$

Therefore, we can see that not all terms are non-negative. Even though,

$$g'(z) \geq 0$$

$$g(z) \geq 0$$

However, the other terms in the above expression may not necessarily be non-negative. For instance, Now, considering the same set of values that I used to show that  $J_1(w)$  is not a convex function,

$$g(z) = 0.5, v_1 = -1, v_2 = 1,$$

For these set of values,

$$v^T H v = (8 - 2)(-2 - 1)(0.25)(0.5)(0.5)$$

$$v^T H v = -18(0.25)(0.25)$$

$$v^T H v = -1.125$$

$$\Rightarrow v^T H v < 0$$

Therefore,  $v^T H v$  is not always non-negative.

$\Rightarrow$  The Hessian matrix  $H$  is not positive semi-definite.

**This implies that  $J_2(w)$  is not a convex function.**

Hence, we know that the **sum operation preserves convexity**. Therefore, I proved that both  $J_1(w)$  and  $J_2(w)$  are not convex functions.

**$\Rightarrow J(w)$  is not a convex function**

B.

We know,

$$J: R^2 \rightarrow R$$

$$J(w) = (\text{logistic}(2w_1 + w_2) - 1)^2 + (\text{logistic}(-2w_1 + w_2) - 0)^2$$

$$\Rightarrow J(w) = (\text{logistic}(2w_1 + w_2) - 1)^2 + (\text{logistic}(-2w_1 + w_2))^2$$

Gradient of  $J$  is given by:

$$\nabla J = \begin{pmatrix} \frac{\partial J}{\partial w_1} \\ \frac{\partial J}{\partial w_2} \end{pmatrix}$$

Firstly, to make derivative calculations easier,

$$\begin{aligned}
& \frac{\partial}{\partial w_1} (\text{logistic}(2w_1 + w_2)) \\
&= \text{logistic}(2w_1 + w_2)(1 - \text{logistic}(2w_1 + w_2)) \frac{\partial}{\partial w_1} (2w_1 + w_2) \\
& \frac{\partial}{\partial w_1} (\text{logistic}(2w_1 + w_2)) = 2 \text{logistic}(2w_1 + w_2)(1 - \text{logistic}(2w_1 + w_2)) \quad (1)
\end{aligned}$$

Similarly,

$$\begin{aligned}
& \frac{\partial}{\partial w_2} (\text{logistic}(2w_1 + w_2)) \\
&= \text{logistic}(2w_1 + w_2)(1 - \text{logistic}(2w_1 + w_2)) \frac{\partial}{\partial w_2} (2w_1 + w_2) \\
& \frac{\partial}{\partial w_2} (\text{logistic}(2w_1 + w_2)) = \text{logistic}(2w_1 + w_2)(1 - \text{logistic}(2w_1 + w_2)) \quad (2)
\end{aligned}$$

Also,

$$\begin{aligned}
& \frac{\partial}{\partial w_1} (\text{logistic}(-2w_1 + w_2)) \\
&= \text{logistic}(-2w_1 + w_2)(1 - \text{logistic}(-2w_1 + w_2)) \frac{\partial}{\partial w_1} (-2w_1 + w_2) \\
& \frac{\partial}{\partial w_1} (\text{logistic}(-2w_1 + w_2)) = -2 \text{logistic}(-2w_1 + w_2)(1 - \text{logistic}(-2w_1 + w_2)) \quad (3)
\end{aligned}$$

Also,

$$\begin{aligned}
& \frac{\partial}{\partial w_2} (\text{logistic}(-2w_1 + w_2)) \\
&= \text{logistic}(-2w_1 + w_2)(1 - \text{logistic}(-2w_1 + w_2)) \frac{\partial}{\partial w_2} (-2w_1 + w_2) \\
& \frac{\partial}{\partial w_2} (\text{logistic}(-2w_1 + w_2)) = \text{logistic}(-2w_1 + w_2)(1 - \text{logistic}(-2w_1 + w_2)) \quad (4)
\end{aligned}$$

**Computing the derivatives for  $J(\mathbf{w})$  with respect to  $\mathbf{w}_1$ :**

$$\begin{aligned}
\frac{\partial J(w)}{\partial w_1} &= \frac{\partial}{\partial w_1} \left( (\text{logistic}(2w_1 + w_2) - 1)^2 + (\text{logistic}(-2w_1 + w_2))^2 \right) \\
\Rightarrow \frac{\partial J(w)}{\partial w_1} &= \frac{\partial}{\partial w_1} ((\text{logistic}(2w_1 + w_2) - 1)^2) + \frac{\partial}{\partial w_1} ((\text{logistic}(-2w_1 + w_2))^2) \\
\Rightarrow \frac{\partial J(w)}{\partial w_1} &= 2(\text{logistic}(2w_1 + w_2) - 1)(2) \frac{\partial}{\partial w_1} (\text{logistic}(2w_1 + w_2)) \\
&\quad + 2(\text{logistic}(-2w_1 + w_2))(-2) \frac{\partial}{\partial w_1} (\text{logistic}(-2w_1 + w_2)) \\
\Rightarrow \frac{\partial J(w)}{\partial w_1} &= 4(\text{logistic}(2w_1 + w_2) - 1) \frac{\partial}{\partial w_1} (\text{logistic}(2w_1 + w_2)) \\
&\quad - 4(\text{logistic}(-2w_1 + w_2)) \frac{\partial}{\partial w_1} (\text{logistic}(-2w_1 + w_2))
\end{aligned}$$

Using (1) and (3),

$$\begin{aligned}
\Rightarrow \frac{\partial J(w)}{\partial w_1} &= 4(\text{logistic}(2w_1 + w_2) - 1) \left( 2\text{logistic}(2w_1 + w_2)(1 - \text{logistic}(2w_1 + w_2)) \right) \\
&\quad - 4(\text{logistic}(-2w_1 + w_2)) \left( -2\text{logistic}(-2w_1 + w_2)(1 - \text{logistic}(-2w_1 + w_2)) \right) \\
\Rightarrow \frac{\partial J(w)}{\partial w_1} &= 8(\text{logistic}(2w_1 + w_2) - 1) \left( \text{logistic}(2w_1 + w_2)(1 - \text{logistic}(2w_1 + w_2)) \right) \\
&\quad + 8(\text{logistic}(-2w_1 + w_2)) \left( \text{logistic}(-2w_1 + w_2)(1 - \text{logistic}(-2w_1 + w_2)) \right) \\
\Rightarrow \frac{\partial J(w)}{\partial w_1} &= -8\text{logistic}(2w_1 + w_2)(1 - \text{logistic}(2w_1 + w_2))^2 \\
&\quad + 8(\text{logistic}(-2w_1 + w_2))^2(1 - \text{logistic}(-2w_1 + w_2))
\end{aligned}$$

**Computing the derivatives for  $J(w)$  with respect to  $w_2$ :**

$$\begin{aligned}
\frac{\partial J(w)}{\partial w_2} &= \frac{\partial}{\partial w_2} \left( (\text{logistic}(2w_1 + w_2) - 1)^2 + (\text{logistic}(-2w_1 + w_2))^2 \right) \\
\Rightarrow \frac{\partial J(w)}{\partial w_2} &= \frac{\partial}{\partial w_2} ((\text{logistic}(2w_1 + w_2) - 1)^2) + \frac{\partial}{\partial w_2} ((\text{logistic}(-2w_1 + w_2))^2)
\end{aligned}$$



$$\Rightarrow \frac{\partial J(w)}{\partial w_2} = 2(\text{logistic}(2w_1 + w_2) - 1)(1) \frac{\partial}{\partial w_2} (\text{logistic}(2w_1 + w_2)) \\ + 2(\text{logistic}(-2w_1 + w_2))(1) \frac{\partial}{\partial w_2} (\text{logistic}(-2w_1 + w_2))$$

$$\Rightarrow \frac{\partial J(w)}{\partial w_2} = 2(\text{logistic}(2w_1 + w_2) - 1) \frac{\partial}{\partial w_2} (\text{logistic}(2w_1 + w_2)) \\ + 2(\text{logistic}(-2w_1 + w_2)) \frac{\partial}{\partial w_2} (\text{logistic}(-2w_1 + w_2))$$

Using (2) and (4),

$$\Rightarrow \frac{\partial J(w)}{\partial w_2} = 2(\text{logistic}(2w_1 + w_2) - 1) \left( \text{logistic}(2w_1 + w_2)(1 - \text{logistic}(2w_1 + w_2)) \right) \\ + 2(\text{logistic}(-2w_1 + w_2)) \left( \text{logistic}(-2w_1 + w_2)(1 - \text{logistic}(-2w_1 + w_2)) \right)$$

$$\Rightarrow \frac{\partial J(w)}{\partial w_2} = -2\text{logistic}(2w_1 + w_2)(1 - \text{logistic}(2w_1 + w_2))^2 \\ + 2(\text{logistic}(-2w_1 + w_2))^2(1 - \text{logistic}(-2w_1 + w_2))$$

Finally, computing the gradient of  $J(w)$ :

$$\nabla J = \begin{pmatrix} \frac{\partial J}{\partial w_1} \\ \frac{\partial J}{\partial w_2} \end{pmatrix}$$

$$\nabla J = \begin{pmatrix} -8\text{logistic}(2w_1 + w_2)(1 - \text{logistic}(2w_1 + w_2))^2 + 8(\text{logistic}(-2w_1 + w_2))^2(1 - \text{logistic}(-2w_1 + w_2)) \\ -2\text{logistic}(2w_1 + w_2)(1 - \text{logistic}(2w_1 + w_2))^2 + 2(\text{logistic}(-2w_1 + w_2))^2(1 - \text{logistic}(-2w_1 + w_2)) \end{pmatrix}$$

## Problem – 4

A.

We know,

$$J(w) = \max \{0, 1 - (2w_1 + w_2)\}$$

Let,

$$J(w) = \max \{J_1(w), J_2(w)\}$$

Such that,

$$J_1(w) = 0$$

$$J_2(w) = 1 - (2w_1 + w_2)$$

→ Consider  $J_1(w)$

We know that a function  $J: R^2 \rightarrow R$  is convex, if for all  $u, v \in R^2$ , and all  $\alpha \in [0, 1]$ ,

$$J((1 - \alpha)u + \alpha v) \leq (1 - \alpha)J(u) + \alpha J(v)$$

Let  $u, v \in R^2$ , and  $\alpha \in [0, 1]$

$$LHS: J_1((1 - \alpha)u + \alpha v) = 0$$

$$RHS: (1 - \alpha)J_1(u) + \alpha J_1(v) = (1 - \alpha) \cdot 0 + \alpha \cdot 0 = 0$$

$$\Rightarrow LHS = RHS = 0$$

$$\Rightarrow J_1((1 - \alpha)u + \alpha v) \leq (1 - \alpha)J_1(u) + \alpha J_1(v) \quad \forall u, v \in R^2 \text{ and } \alpha \in [0, 1]$$

Therefore,  $J_1(w)$  is a convex function.

→ Consider  $J_2(w)$

We know that a function  $J: R^2 \rightarrow R$  is convex, if for all  $u, v \in R^2$ , and all  $\alpha \in [0, 1]$ ,

$$J((1 - \alpha)u + \alpha v) \leq (1 - \alpha)J(u) + \alpha J(v)$$

$$J_2(w) = 1 - (2w_1 + w_2)$$

$$\Rightarrow J_2(w) = 1 - 2w_1 - w_2$$

$$\Rightarrow J_2(w) = -2w_1 - w_2 + 1$$

In matrix form,

$$\Rightarrow J_2(w) = \begin{pmatrix} -2 & -1 \end{pmatrix} \cdot \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} + 1$$

$$\Rightarrow J_2(w) = Aw + b$$

where,

$$A = \begin{pmatrix} -2 & -1 \end{pmatrix}$$

$$w = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}, \quad w \in R^2$$

$$b = 1$$

We know that a function  $J: R^2 \rightarrow R$  is convex, if for all  $u, v \in R^2$ , and all  $\alpha \in [0,1]$ ,

$$J((1 - \alpha)u + \alpha v) \leq (1 - \alpha)J(u) + \alpha J(v)$$

Let  $u, v \in R^2$ , and  $\alpha \in [0,1]$

$$LHS: J_2((1 - \alpha)u + \alpha v)$$

$$= A((1 - \alpha)u + \alpha v) + b$$

$$= A(u - \alpha u + \alpha v) + b$$

$$= Au - \alpha Au + \alpha Av + b$$

$$RHS: (1 - \alpha)J_2(u) + \alpha J_2(v)$$

$$= (1 - \alpha)(Au + b) + \alpha(Av + b)$$

$$= Au + b - \alpha Au - \alpha b + \alpha Av + \alpha b$$

$$= Au - \alpha Au + \alpha Av + b$$

$$\Rightarrow LHS = RHS$$

$$\Rightarrow J_2((1 - \alpha)u + \alpha v) \leq (1 - \alpha)J_2(u) + \alpha J_2(v) \quad \forall u, v \in R^2 \text{ and } \alpha \in [0,1]$$

Therefore,  $J_2(w)$  is a convex function.

Thus, we know that the max operation preserves convexity,

for  $J_1: R^2 \rightarrow R$ , and  $J_2: R^2 \rightarrow R$

$$J(w) = \max\{J_1(w), J_2(w)\}$$

Since,  $J_1(w)$  and  $J_2(w)$  are convex functions,

**$J(w)$  is a convex function.**

**B.**

Our goal is to find an affine function  $A(w)$  that satisfies the following conditions:

$$J(u) = A(u), \text{ and } J(w) \geq A(w) \quad \forall w \in R^2 \quad \text{at } u = \left(0, \frac{1}{2}\right)$$

So, the affine predictor is given by:

$$A(w) = J(u) + \nabla J(u)^T (w - u)$$

We know,

$$J(w) = \max \{0, 1 - (2w_1 + w_2)\}$$

**CASE-1:**  $1 - (2w_1 + w_2) > 0$

$$\Rightarrow J(w) = 1 - (2w_1 + w_2)$$

Computing partial derivatives of  $J(w)$ ,

$$\frac{\partial J}{\partial w_1} = -2$$

$$\frac{\partial J}{\partial w_2} = -1$$

$$\nabla J = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

Hence,  $\nabla J$  at  $u = \left(0, \frac{1}{2}\right)$ :

$$\nabla J = \begin{pmatrix} -2 \\ -1 \end{pmatrix} \tag{1}$$

Hence,

$$J(u) = 1 - (2u_1 + u_2) \tag{2}$$

$$A(w) = J(u) + \nabla J(u)^T (w - u)$$

Using (1) and (2),

$$A(w) = 1 - (2u_1 + u_2) + \begin{pmatrix} -2 \\ -1 \end{pmatrix}^T \begin{pmatrix} w_1 - u_1 \\ w_2 - u_2 \end{pmatrix}$$

$$A(w) = 1 - (2u_1 + u_2) + (-2 \quad -1) \begin{pmatrix} w_1 - u_1 \\ w_2 - u_2 \end{pmatrix}$$

$$A(w) = 1 - (2u_1 + u_2) - 2(w_1 - u_1) - (w_2 - u_2)$$

**CASE-2:**  $1 - (2w_1 + w_2) \leq 0$

$$\Rightarrow J(w) = 0 \quad (3)$$

Computing partial derivatives of  $J(w)$ ,

$$\frac{\partial J}{\partial w_1} = 0$$

$$\frac{\partial J}{\partial w_2} = 0$$

$$\nabla J = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (4)$$

$$A(w) = J(u) + \nabla J(u)^T (w - u)$$

Using (3) and (4),

$$\Rightarrow A(w) = 0$$

Therefore, the overall affine function becomes:

$$A(w) = \max\{0, 1 - (2u_1 + u_2) - 2(w_1 - u_1) - (w_2 - u_2)\}$$

Thus, using the above two cases that we discussed:

**When  $1 - (2w_1 + w_2) > 0$ ,**

$$J(w) = 1 - (2w_1 + w_2)$$

$$A(w) = 1 - (2u_1 + u_2) - 2(w_1 - u_1) - (w_2 - u_2)$$

$$\Rightarrow A(w) = 1 - 2u_1 - u_2 - 2w_1 + 2u_1 - w_2 + u_2$$

$$\Rightarrow A(w) = 1 - 2w_1 - w_2$$

$$\Rightarrow A(w) = J(w)$$

**When  $1 - (2w_1 + w_2) \leq 0$ ,**

$$J(w) = 0$$

$$A(w) = 0$$

$$\Rightarrow A(w) = J(w)$$

Therefore,

$$A(w) \geq J(w) \quad \forall w \in R^2$$

$$\text{At } u = \left(0, \frac{1}{2}\right),$$

$$J(u) = \max\{0, 1 - (2u_1 + u_2)\}$$

$$J(u) = \max\{0, 1 - \left(2(0) + \frac{1}{2}\right)\}$$

$$J(u) = \max\{0, \frac{1}{2}\}$$

$$J(u) = \frac{1}{2}$$

Also,

$$A(u) = \max\{0, 1 - (2u_1 + u_2) - 2(u_1 - u_1) - (u_2 - u_2)\}$$

$$A(u) = \max\{0, 1 - (2u_1 + u_2)\}$$

$$A(u) = \max\{0, 1 - \left(2(0) + \frac{1}{2}\right)\}$$

$$A(u) = \max\{0, \frac{1}{2}\}$$

$$A(u) = \frac{1}{2}$$

$$\Rightarrow A(u) = J(u)|_{u=(0, \frac{1}{2})}$$

Therefore,

$$A(w) = \max\{0, 1 - (2u_1 + u_2) - 2(w_1 - u_1) - (w_2 - u_2)\}$$

Satisfies:

$$\begin{aligned} A(w) &\geq J(w) \quad \forall w \in R^2, \text{ and} \\ &\Rightarrow A(u) = J(u) \Big|_{u=(\frac{1}{4}, \frac{1}{2})} \end{aligned}$$

**C.**

Our goal is to find an affine function  $A(w)$  that satisfies the following conditions:

$$J(u) = A(u), \text{ and } J(w) \geq A(w) \quad \forall w \in R^2 \quad \text{at} \quad u = \left(\frac{1}{4}, \frac{1}{2}\right)$$

So, the affine predictor is given by:

$$A(w) = J(u) + \nabla J(u)^T(w - u)$$

We know,

$$J(w) = \max\{0, 1 - (2w_1 + w_2)\}$$

**CASE-1:**  $1 - (2w_1 + w_2) > 0$

$$\Rightarrow J(w) = 1 - (2w_1 + w_2)$$

Computing partial derivatives of  $J(w)$ ,

$$\frac{\partial J}{\partial w_1} = -2$$

$$\frac{\partial J}{\partial w_2} = -1$$

$$\nabla J = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

Hence,  $\nabla J$  at  $u = \left(\frac{1}{4}, \frac{1}{2}\right)$ :

$$\nabla J = \begin{pmatrix} -2 \\ -1 \end{pmatrix} \tag{1}$$

Hence,

$$J(u) = 1 - (2u_1 + u_2) \tag{2}$$

$$A(w) = J(u) + \nabla J(u)^T(w - u)$$

Using (1) and (2),

$$A(w) = 1 - (2u_1 + u_2) + \begin{pmatrix} -2 \\ -1 \end{pmatrix}^T \begin{pmatrix} w_1 - u_1 \\ w_2 - u_2 \end{pmatrix}$$

$$A(w) = 1 - (2u_1 + u_2) + (-2 \quad -1) \begin{pmatrix} w_1 - u_1 \\ w_2 - u_2 \end{pmatrix}$$

$$A(w) = 1 - (2u_1 + u_2) - 2(w_1 - u_1) - (w_2 - u_2)$$

**CASE-2:**  $1 - (2w_1 + w_2) \leq 0$

$$\Rightarrow J(w) = 0 \quad (3)$$

Computing partial derivatives of  $J(w)$ ,

$$\frac{\partial J}{\partial w_1} = 0$$

$$\frac{\partial J}{\partial w_2} = 0$$

$$\nabla J = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (4)$$

$$A(w) = J(u) + \nabla J(u)^T (w - u)$$

Using (3) and (4),

$$\Rightarrow A(w) = 0$$

Therefore, the overall affine function becomes:

$$A(w) = \max\{0, 1 - (2u_1 + u_2) - 2(w_1 - u_1) - (w_2 - u_2)\}$$

Thus, using the above two cases that we discussed:

**When  $1 - (2w_1 + w_2) > 0$ ,**

$$J(w) = 1 - (2w_1 + w_2)$$

$$A(w) = 1 - (2u_1 + u_2) - 2(w_1 - u_1) - (w_2 - u_2)$$

$$\Rightarrow A(w) = 1 - 2u_1 - u_2 - 2w_1 + 2u_1 - w_2 + u_2$$

$$\Rightarrow A(w) = 1 - 2w_1 - w_2$$

$$\Rightarrow A(w) = J(w)$$



**When  $1 - (2w_1 + w_2) \leq 0$ ,**

$$J(w) = 0$$

$$A(w) = 0$$

$$\Rightarrow A(w) = J(w)$$

Therefore,

$$A(w) \geq J(w) \quad \forall w \in R^2$$

$$\text{At } u = \left(\frac{1}{4}, \frac{1}{2}\right),$$

$$J(u) = \max\{0, 1 - (2u_1 + u_2)\}$$

$$J(u) = \max\{0, 1 - \left(2\left(\frac{1}{4}\right) + \frac{1}{2}\right)\}$$

$$J(u) = \max\{0, 0\}$$

$$J(u) = 0$$

Also,

$$A(u) = \max\{0, 1 - (2u_1 + u_2) - 2(u_1 - u_1) - (u_2 - u_2)\}$$

$$A(u) = \max\{0, 1 - (2u_1 + u_2)\}$$

$$A(u) = \max\{0, 1 - \left(2\left(\frac{1}{4}\right) + \frac{1}{2}\right)\}$$

$$A(u) = \max\{0, 0\}$$

$$A(u) = 0$$

$$\Rightarrow A(u) = J(u)|_{u=\left(\frac{1}{4}, \frac{1}{2}\right)}$$

Therefore,

$$A(w) = \max\{0, 1 - (2u_1 + u_2) - 2(w_1 - u_1) - (w_2 - u_2)\}$$

Satisfies:

$$A(\mathbf{w}) \geq J(\mathbf{w}) \quad \forall \mathbf{w} \in \mathbb{R}^2, \text{ and}$$

$$A(\mathbf{u}) = J(\mathbf{u})|_{\mathbf{u}=(\frac{1}{4}, \frac{1}{2})}$$

### Problem – 5

**Frobenius Norm Difference:** 1.7813461687902673e-11

This is quite a small number, so it seems that the matrices are very similar.

### Problem – 6

**A. Euclidean norm of the gradient at  $(\mathbf{w}_{mle}, \mathbf{b}_{mle})$ :** 1.9359452831386658e-10

**B.**

**Training Error Rate:** 0.0453

**Test Error Rate:** 0.0540

### Problem – 7

The balance error rate takes the shape:

$$\frac{1}{2} \Pr(f(X) = 1 \mid Y = 0) + \frac{1}{2} \Pr(f(X) = 0 \mid Y = 1)$$

But  $\Pr(f(X) = 1 \mid Y = 0)$  is always 0 because the classifier always returns 0, so the balance error rate in our case becomes just:

$$\frac{1}{2} \Pr(f(X) = 0 \mid Y = 1)$$

Now,

$$\frac{1}{2} \Pr(f(X) = 0 \mid Y = 1) = \frac{1}{2} \Pr(f(X) = 0) \Pr(Y = 1)$$

This is because the two events are independent of each other, as:

$$Pr(f(X) = 0) = 1, \text{ and } Pr(Y = 1) = [\textit{fraction of data where } Y = 1]$$

And, therefore, this simply provides us our original error rates from the ones calculated in Problem-6 divided by a factor of 2.

**Balanced Training Error Rate:** 0.0227

**Balanced Test Error Rate:** 0.0270

## **Problem – 8**

**Balanced Training Error Rate:** 0.4959

**Balanced Test Error Rate:** 0.5790